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Virginia Board of Education, Special Meeting, Tuesday, May 21, 2024

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- Official meeting video (VDOE, YouTube): https://www.youtube.com/watch?v=cdFJQWt6NHw
- Board of Education meetings, agendas and minutes (VDOE): https://www.doe.virginia.gov/state-board-data-funding/virginia-board-of-education

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[00:06:57] 08.3 B 3 of the code of Virginia, because his principal residence is more than 60 miles from the meeting location, and so we will proceed. So what I would like to do at this time is have a motion to approve or amend the agenda as presented.

[00:07:25] move item c to the first item on our agenda in front of a and b i think it's important to have the update before we move forward on the two proposals i have a second on that motion okay all in favor indicate by saying aye aye thank you so we move item c up to item a and so that brings uh miss dr armstrong to us president yeah if you're here i'll sit if that's okay

[00:08:05] uh thank you good morning madam president board members and dr raley i would like to provide the board with a brief overview of the currently obligated college partnership lab school grant funds on behalf of the 12 board approved lab schools that's okay no problem you just let me

[00:08:32] Let me know when it comes. So we'll be in our public comment today.

[00:08:47] You would probably like to remind everyone that we have a great problem and respect with them. Thank you, and respect for others. Please share your thoughts today. Respect for the meetings of the week. At the end, please provide your name and your role.

[00:09:05] We believe that all remarks might be relevant to the chair, not to any other individual Any quick comments should be provided to the chat and the back screen, and the speaker will be in three minutes of shared remarks.

[00:09:22] There will be a timer on the screen, it will get started in three minutes, and then we will get a yellow screen as we get to count down to students, and then the screen will for me to build our values. Again, comments go to Mr. Chapman, or you may also email your comments to bome at bome.brirginia.gov. And we only have two people signed up today. So

[00:09:54] we will speak with Karen and Ronald Jones, probably with I'm spinning more. Thank you.

[00:10:35] Good morning. Thank you for allowing me the opportunity to speak to you today. My name is Dr. Erin McDonnell-Jones, and I'm an associate professor of teacher education at Randolph-Macon College in Ashland, Virginia. I joined Randolph-Macon last fall, and it has brought immeasurable joy to work with these faculty members and students over the past year.

[00:10:55] During my time, I have been called upon by many students to help them prepare for and pass the Virginia Communication and Literacy Assessment, otherwise known as the VCLA. They would be here with me today to speak with you, but this is our finals week, So I hope you will hear me knowing that I speak for many of them.

[00:11:14] These students are smart, hardworking, focused, and determined to become the best teacher that they can be for their future Virginia students. Unfortunately, they also have to deal with multiple issues associated with standardized testing. The financial cost of these tests add up for multiple times students are required to take them.

[00:11:32] The tests perpetuate educational inequalities through a biased design. And most importantly, the tests do not predict who will or will not be a good teacher. As you know, on April 8th of this year, the governor approved HB 731, which eliminates the requirements for teachers to pass the VCLA.

[00:11:50] The legislation requires that the Board of Education take action to determine alternative methods of assessment. I'm here today to urge you to take action soon in order to allow our students to continue towards licensure in the next academic year. I can assure you that these students are able to meet the objectives of what the VCLA measured, but there are better ways to demonstrate it, such as through collegiate coursework, project-based assessments, and performance-based tasks.

[00:12:21] The VCLA does not quantify the true ability of our future teachers. Unfortunately, if the board does not take action soon, we will lose out on licensing many future teachers who will enter the classroom fully prepared for their teaching career. Again, I urge you to take action soon allow our students to continue towards licensure and demonstrate their communication and literacy

[00:12:43] skills through alternative needs thank you thank you sydney waldrop thank you um good morning and thank you for allowing me the opportunity to speak to you today my name is sydney waldrop and

[00:13:18] i'm a sophomore at randolph making college i'm a double major in history and exceptional education with a minor and secondary education with the goal of being a middle school history teacher in second grade i was diagnosed with dyslexia and ever since then i've made it my life's mission to become a teacher so that i can help students with learning disabilities and all students reach their full potential no

matter what differences they may have i have come to you today to talk to you about the virginia communication and literacy assessment otherwise known as the vcla

[00:13:56] The VCLA has affected me personally because I have taken it and been unable to pass. This test has affected me emotionally because it has continued to put a lot of stress and anxiety on me throughout my school year. As a student with dyslexia, this test can be hard for me to pass because it focuses on topics that I struggle with.

[00:14:18] Because I have not passed the VCLA, I was unable to be accepted into the teacher preparation program with the rest of my cohort this spring. I know that I'm going to be an excellent teacher, but this has caused me to feel left out due to the requirement from the Department of Education.

[00:14:39] On April 8th of this year, the governor approved House Bill 731, which eliminates the requirements for teachers to pass the VCLA. The legislation requires the Board of Education to take action to determine alternate methods of assessment.

[00:14:54] I'm here today to ask you to determine what alternate formats would be better and more inclusive to measure future education students' communication and literacy. Through coursework, students can show communication and literacy through project-based learning and teacher prep classes and other classes.

[00:15:18] The VCLA does not measure my teaching ability. It puts emphasis on a learning disability that I've been diagnosed with for most of my life. My disability does not define me and neither should my test score on the VCLA. Unfortunately, if the board does not take action soon, this could ultimately impact my future. It may mean that I never get to teach students in the classroom like I've always dreamed of doing. Thank you for your time.

[00:15:48] If there are any other speakers that wish to speak today, please line up behind the microphone. Okay, I'm going to close public.

[00:16:05] And we do, before Dr. Armstrong comes up, we do need to approve the minutes from our meeting as were presented in SharePoint. Do I have a motion on those? Ms. Ashton?

[00:16:24] Aye. Dr. Northern? Mr. Hanson? Mr. Rotherham? Aye. Dr. Siebert? Aye. Ms. Holton? Aye. And Chair votes aye. Mr. Rotherham, can you hear us clearly? I just want to make sure.

[00:16:39] I can hear you. I heard you live. No, yes, I did. I just want to make sure that we're loud and clear for you. Thank you. Okay, Dr. Armstrong. Thank you, Madam President and board members.

[00:16:57] Again, I'd like to provide you all, the board, Dr. Raley, with a brief overview of the currently obligated College Partnership Lab School Grand Funds on behalf of the 12 approved lab schools that the board has approved since July.

[00:17:15] And as soon as I get the graphic up, I will walk you through that. I know, yeah. I'll honor my time limit as well, members of the board. Thank you. There we go.

[00:17:36] This graph is aligned with the language of the FY 2024 budget caboose bill, which limits the overall grant fund to $75 million. And as you can see, as of today, prior to consideration of the two lab school applications before the board, the total obligated funds for the 12 board-approved lab schools totals approximately $64.5 million. This includes awarded planning grant funds, startup funds, and operational

funds. In full transparency, the University of Virginia lab school contract is out for signature as I speak, and we anticipate full execution of that contract this morning. If it has

[00:18:14] not already been completed, I'm glancing back at Ms. McDaniel, who's keeping me updated that by the minute. You will note that a full two-thirds of the obligated funds are designated for four-year public university lab schools, and those include James Madison, Mary Washington, University of Virginia, George Mason, Virginia Commonwealth University, and both the ODU Newport News and ODU Chesapeake lab schools. Our two-year public lab schools include Germana Community College, Mountain Gateway Community College, and Paul D. Camp, which account for nearly $13 million, while the two approved private lab schools represent less than 10 of the $64.5 million

[00:18:53] dollars of obligated funds as we sit here today. Looking forward, if you see to the right in that box, today's lab score applications for consideration represent funding requests of slightly more than seven million dollars, and if approved and contracted, the balance in the fund would sit at nearly 3.5 million. Several applications do remain in various stages of the approval process currently, and we anticipate at least two applications for review at the June standing committee meeting details of that are being finalized and as a brief overview of where we are with the fund i'm happy to to take any questions are there any questions for dr armstrong

[00:19:32] please miss holton uh thank you uh dr armstrong this is very helpful the potential fund balance if both of the applications before us today are approved the fund balance would be three and a half million am i understanding that correctly yes ma'am and and you anticipate that we may have as many as two applications coming before us in june norfolk state and another odu potentially so do we know what the amount of those applications is we do not at this time uh we're still reviewing

[00:20:08] those applications our subject matter experts so do we know what they've applied for the amount they've applied for we're reviewing that currently i don't have the numbers handy but um we're considering that with our overall budget so uh okay assuming that those applications are in the same range as the applications we've seen before which we're all i think i don't think we've seen any the george mason frederick at 2.8 million may be the the lowest number we've

[00:20:40] seen uh if we in fact do get two more applications we may not be able to fund two applications that two more that succeed in making it to the board we may not be able to fund all the applications that are coming before us am i correct assuming that those two more are in the same financial range in addition to the the numbers that we have in the fund there are also some remaining questions that i know we'll discuss regarding fiscal agency for some of our two-year and private schools as well we we have to finalize that that doesn't necessarily translate to additional funds however there are several variables i think are still in play that we we have to monitor in the weeks to

[00:21:20] come so it's possible that some of the funds that have already some of the programs that have already been approved uh may the fund some of those funds conceivably could come back into the hopper we we hope not but um that of course is a possibility okay i'm just recognizing that at least one of the two applications you've got coming up is from a school that we haven't yet seen whereas the applications that are before us today are from universities that have already had approved applications so it's true i just i'm looking forward to our especially our committee

[00:21:57] members mr rotherham and mr hanson who've been so intimately involved in this helping us think through what do we need to do to make sure we have the opportunity to fund good programs that

come before us in june uh if any so help us think that through mr hanson miss holton and i just uh do think that we really need to we've had a lot of uncertainty kind of for us we've had a lot of uncertainty and change thrust upon us here in the last week or two and i think we owe it both to the

[00:22:32] lab committee to be about our work at our june 3rd meeting as well as the departmental staff but also just with some of the other stipulations that were put in the legislation all relatively new and I think the department and the schools have some work to do, frankly, and so I think we owe it to the schools to honor the process, and we have another lab school meeting coming up in 10 days or whatever it is, and another board meeting coming up here in June, so I think we, with the

[00:23:08] June 30th deadline for all of this, so I think we just, I went to the schools to just work through the process here in the coming weeks. Any other questions or comments at this time? Okay.

[00:23:30] I would like to move us from, I think because there are some legal questions around how the board iterates from here, that we might want to go ahead and take this opportunity to move into our closed session now, if that's the will of the board to do that prior to taking up the two lab school applications.

[00:23:57] I would move that the Virginia Board of Education convene in a closed meeting for the following purpose, consultation with legal counsel employed by or attained by a public body regarding specific legal matters requiring a provision of legal advice by such counsel pursuant to Section 2.2-3711A8 of the Code of Virginia, and specifically related to college partnership laboratory schools, and the legal counsel to the Virginia Board of Education staff whose presence will aid in the Board's consideration of

[00:24:31] this matter participate in the closed meeting second and we will get into closed session now sorry sorry hold on miss ashton miss kilgore is not here dr northern mr hanson hi mr rotherham aye dr siebert miss holton and chair votes aye okay and we are going into closed session

[00:25:04] thank you closed meeting pursuant to a recorded vote on the motion above and in accordance with

[01:42:56] the freedom of information act section 2.2-3712 of the code of virginia requires that the board certify that to the best of each member's knowledge only public business matters lawfully exempt from open meeting requirements under this chapter and only such public business matters as were identified in the motion by which the closed meeting was convened were heard discussed or considered any member who believes there was a departure from these requirements shall so state prior to the vote, indicating the substance of the departure that in his or her judgment has taken place. The statement of the departure will be recorded in the minutes. So moved.

[01:43:37] Dr. Northern, hi. Hi. All right. Now we will move on to item B, or I guess, yeah, I guess it is item B, huh?

[01:44:00] Officially. Thank you. Madam President and board members, first, thank you all for convening today's special meeting to include consideration of two lab school applications. On behalf of the VDOE, I would like to provide an overview of the Old Dominion University STEM Academy at Booker T.

[01:44:30] Washington Elementary School lab school proposal, a partnership between Old Dominion University and Suffolk City Public Schools. I would like to first thank the members of the College Partnership Lab School Standing Committee, including Chair Mrs. Joan Wadisca, Virginia Board of Education members Mr. Bill Hanson, Mr. Andy Rotherham, and members Mr. John Bailey and Dr. Pam

Moran for their time, feedback, and commitment to lab schools and additional opportunities afforded by lab schools on behalf of the students of the Commonwealth.

[01:44:57] Upon receipt of a lab school application, the VDOE convenes a subject matter expert team to provide detailed feedback in each compliance-related section of the application. The goal of the Application Review Technical Expert Committee is to document the needed technical expertise and provide counseling to the applicants to identify and correct any potential deficiencies in the planning and implementation of the proposed College Partnership Laboratory School, determine compliance with state law and regulation, highlight potential problems or red flags with information contained in the application, and to identify aspects of the proposal that are especially exciting or unique.

[01:45:31] The subject matter expert team includes me, Ms. Shawna LeBlanc from the Office of Licensure, Mr. Jason Ellis from the Office of Accountability, Ms. Rebecca Askew from the Office of Policy, Mr. Ed Lancer from the Office of Budget, Ms. Melissa Velasquez from the Department of Policy and Communications, Dr. Samantha Hollins from the Department of Special Education Student Services, Dr. Dave Myers from Data Research and Technology, Ms. Claire Bremner, Innovation Coordinator, and Ms. Sophia McDaniel from the Office of Innovation.

[01:45:59] Each subject matter expert was provided with a laboratory school criteria checklist worksheet, and areas of expertise were assigned to various staff subject their education expertise at the VDOE. A compilation of these comments is included in the Excel spreadsheet entitled VDOE Completeness and Compliance Report.

[01:46:15] The completeness and compliance report corresponds to the lab school application and reflects the following dimensions of compliance. Executive summary, mission and vision, educational program, governance, management structure, financial and operations, placement plan, and other assurances. The application presented to you today has undergone multiple phases of subject matter review.

[01:46:34] And additionally, the agency has engaged the Office of the Attorney General and the Governor's Transformation Office to ensure that applications presented to the Standing Committee and Board of Education are complete and compliant and that budget and sustainability plans are scrutinized for efficiency and effectiveness in the use of lab school grant funds and the Higher Ed Partners detailed plans to sustain the lab school past the grant fund window.

[01:46:55] You will find cover sheets from the OAG and VDOE as well as consistent budget templates in each application reflecting these additional efforts taken on behalf of the applications before you today for consideration, including the ODU Suffolk application. This application was presented to the Lab School Standing Committee on May 6, 2024, and was unanimously approved on first and final review to be forwarded for your consideration today.

[01:47:18] Joining me today from Old Dominion University and Suffolk City Public Schools are Dr. Akima Branch, Chief Academic Officer of Suffolk City Public Schools. Dr. John Gordon, who was with us and had to slip away for his daughter's event. I hope he comes back, who's the Superintendent of Suffolk Public Schools. Dr. Caitlin Leitner, who's the Coordinator of Instruction for Suffolk Public Schools.

[01:47:37] Dr. Tish Zemerski, Associate Vice President of the Regional Higher Education Center for Old Dominion University, and Dr. Karen Sanzo is joining us online from Old Dominion University. Upon

completion of my report, I will invite the ODU team to provide a brief overview of their proposal and to introduce any other special guests representing the STEM Academy at Booker T. Washington Elementary.

[01:47:56] The proposed lab school named ODU STEM Academy is an innovative, interdisciplinary lab school model that focuses on providing career exposure to students through real-world problem-solving in collaboration with higher education professionals, pre-service college students, and industry experts. The goals of the ODU STEM Academy include student engagement and high-level application of real-world problem-solving while working closely with a team of experts from both industry and higher ed partners.

[01:48:22] The provision of a space for innovation where educators, industry, and pre-service professionals can explore best practices in teaching and learning and student engagement coursework aligned with personal interests and regional market demand industry and business, streamlining the pipeline for high-wage, high-skill, high-demand employment in the region served by the lab school.

[01:48:40] Students will remain enrolled in their home school division while being provided with several options to engage in career exploration, including innovative hands-on elementary STEM curriculum. I will also note that this is our first, and as far as I know, only lab school application from the elementary level as well.

[01:48:57] This early introduction to STEM is coupled with engaging opportunities for students supported by a wide network of community supporters. The proposed lab school will serve 120 students from grades K through 5. Instruction will be administered in a hands-on environment and include simultaneous teaching by SPS staff, ODU faculty and pre-service students, and industry experts.

[01:49:16] The VDOE received the ODU STEM Academy application on February 1, 2024, and in the intervening months, we have collaborated with the applicant as follows. The applicant was responsive to SME feedback and questions and provided edits. Their application and addendum is required to meet compliance.

[01:49:33] Prior to the Standing Committee's first and final review on May 6, the VDOE staff provided technical assistance to the ODU STEM Academy team in preparation for review. Prior to their submission, technical assistance calls were held between VDOE and the STEM Academy teams on September 11, 2023, December 4, 2023, and January 8, 2024.

[01:49:52] Following submission of the application, additional technical assistance calls were held on April 17, 2024. The first round of SME reviews provided the week of March 12, 2024, and a revised application was submitted to VDOE on March 25. A revised application, budget, and sustainability agenda were also provided to the VDOE on April 17, 2024, and a final application responsive to suggested revisions was received on April 24 for consideration by the Standing Committee.

[01:50:20] Follow-up feedback has been provided to the applicants. The ODU STEM Academy team presented to the Standing Committee on May 6, responded to questions of the committee, and was approved by the Standing Committee to proceed to today's special meeting of the Board of Education for consideration for approval. The ODU STEM Academy application was posted for public comment following the May 6th Standing Committee meeting on May 6th and remained available until May 20th, and there was no comment received.

[01:50:45] On behalf of the VDOE, I submit that the Old Dominion University STEM Academy application is compliant and ready to implement, and I recommend to the Board that the labs will be approved and

subsequently begin the contract approval process. Here to share more again about ODU STEM Academy is Dr. Akima Branch, Dr. Gordon, Dr. Leitner, Dr. Zemerski, and Dr. Sanzo, who will also introduce additional guests and provide additional information and answer any questions from the board.

[01:51:11] Good morning, Virginia, Board of Education. We are pleased and honored to have made it to this point in the STEM Lab Academy application journey.

[01:51:32] As Dr. Armstrong stated, I am Dr. Okima Branch, Chief Academic Officer of Suffolk Public Schools, joined by my esteemed colleagues. We will not belabor this moment and are ready to proceed. We presented the STEM Academy at Booker T. Washington to the committee, and we were very proud of our work with the Virginia Department of Education and all that it took working in partnership with Old Dominion University to get us to this point

[01:52:07] and to see our vision for our STEM Academy in progress and hopefully to come to fruition. Our mission is simple, but yet complicated. Seems like the world of education today.

[01:52:25] Our mission supports excellence in education. And as Dr. Armstrong mentioned, we are the only elementary STEM lab academy or application that has been presented. We want to start earlier than waiting for secondary education to really expose students to a curriculum that focuses them on STEM education and careers. We want to broaden our students' horizons and inspire them, I guess I do have my notes in front of me, with a sense of instilled sense of responsibility. We are going to

[01:53:02] be using a design thinking framework where we will have more student agency and voice in their learning as they will be entering a world of problems that they will solve based on their level and where they see things happening in our world, in our community, especially in the schools.

[01:53:26] We will address the needs of low socioeconomic populations. Booker T. Washington Elementary School is one of our elementary schools that has the lowest socioeconomic status. We thought in a short way about the academy being at Booker T. Washington to increase the level of exposure to STEM, not only to that school and that community but also to students so that they are exposed at an earlier age to STEM in ways that they may not see it on a day-to-day basis in their communities.

[01:54:04] Provide an innovative hands-on experiential dynamic teaching and learning lab for students to explore the world of STEM. What's very important here is that there will be an exploratory setting daily. Students will not come in and be in a traditional model. Yes, we will have a standard schedule, but Dr. Leitner will explain what that day in the life of a student looks like.

[01:54:31] It will be very engaging. They'll explore their learning and be able to present it to not only the school and their teachers, but also our community. We want to expand the partnerships also that we already have so that they see our students starting early on solving some of their own issues. And then we will be collaborating with Old Dominion University for that integration of teaching and learning and practice to build best practice for our staff to be better implementers of a design thinking model and instruction, STEM instruction for our students. Our overall vision,

[01:55:11] we're cultivating a generation of productive young citizens who are prepared for our global world. They'll have a greater awareness, creative capacity, understanding skills, and knowledge needed for their success throughout their educational experiences. Again, we're starting very early. College and STEM careers, our community, our society, and beyond. I'm going to turn it over to Dr. Caitlin Lightner to review

our instructional model. Perfect. Thank you so much. Good morning to all. I am ecstatic to be here to present this to you all. As Dr. Branch stated, we need to catch

[01:55:51] them early. As I oversee K-12 STEM education in our city, I have noticed a need to catch our students much younger than at the high school level. Perfect. Thank you so much. I'll use my teacher voice. So excited about this opportunity to start with kindergarten all the way up through fifth grade. For our instructional model, we are looking at design thinking. So this is taking our students to the 5E learning model that's adopted by the VDOE. So we'll keep that in our curriculum for our overview. But our goal is to expose students to local issues around our community,

[01:56:27] around the Commonwealth, around the state of Virginia. And our goal, our overall goal and purpose is to produce critical thinkers and problem solvers. So while we will still abide by the Virginia standards of learning, we will be doing it in a different way through the design thinking way. Students will be given an audience of who they're trying to communicate to in the problem that they're in which they're trying to solve. So a day in the life of a lab school student could look as if the playground doesn't meet the needs of all of our community members. Grandparents day, they're not all able to come due to wheelchair crutches, et cetera. And so they are given that design thinking process where they have to problem solve. Their math class is going to look like

[01:57:05] budgets. They have to come up with a budget because everything has a price. They're going to do research. So they're going to be reading and writing about the research of what their proposal is. Of course, the science is that STEM integration. So you have your force motion and energy connection there. And of course, the social studies, you have to know about your city in your Commonwealth. So every day would look different. Every grade level would look different. But the purpose is to produce problem solvers for a 21st century world. And we couldn't do this without our partners and supporters and we have many um these are just a few and many others have offered

[01:57:41] letters of support but oh but this this list will continue to grow thank you one of the i think the most exciting partnerships we have um has actually started about a year ago with uh suffolk public schools and odu um we met uh simply to talk about ways to work together and we had no idea we'd be sitting here today so we launched a series at odu of four workshops that extend the project lead the way curriculum in the 11th grade classrooms in suffolk schools and they've been visiting these

[01:58:16] students have been visiting odu tri-city center for the last four months it will culminate in june but we've also partnered with the hampton roads biomedical research consortium and they are They're providing in these workshops demonstrations in the 3D lab that are then followed up with real world engagement with ODU faculty as well as industry business and industry leaders on what has been learned in the lab.

[01:58:43] And so due to the success of this practical application and the network that this is building for our 11th graders, we already started the development of two programs for next year with Hampton Roads, and for 11th and 12th graders. So we're very excited about that. But of course, we will continue to grow this list and also use some of our internal partners at ODU through ODU STEM committees and related groups and organizations. Sustainability plan. So our partners will help us with our sustainability. ODU has a strong history of sustainability with its

[01:59:20] initiatives, many of which have already started, and commitments from Suffolk Public Schools and ODU. We intend to pursue grants, leverage our research to demonstrate the innovations we wish to

continue as we move into our self-sustaining years, and we will leverage our educator prep program, our Monarch Teach program, that has already had a decade of grant funding. ODU's innovation lab has already received funding to bring the go tech teaching lab to life so we'll be using that as well as ongoing collaboration with hampton roads uh workforce council to stay

[01:59:57] abreast of any stem related initiatives um in the region but i want to mention two um two initiatives that i think uh are critical to um to what we're all doing here with these lab schools one is to engage ODU's Department of Development and Advancement because they will help us leverage our alumni and business and industry leaders and help us create a campaign for fundraising that's somewhat unique and adds to the wonderful complexity of what we're doing. But that will

[02:00:32] bring a lot of us together and a lot more support for our sustainability. And related to that, as you know, ODU has also put forth two other proposals for lab schools, and we would really enjoy leveraging those capabilities, celebrating the unique aspects of each school, as well as the similarities and the shared resources, for example, of what we might find with those.

[02:00:57] So there's a lot of opportunity with the way that we're bringing forth this particular lab school. And to close us out, our outcomes. We believe we have articulated our overall mission and vision and purpose for this elementary lab school. We'll have a hands-on practical STEM approach to our students' learning.

[02:01:21] There will be daily instruction where, because this is K-5, it will be cross-curricular. The learning will have a STEM foundation, but we will cover everything that our students are expected to have in a K-5 environment.

[02:01:39] It will enhance that student academic proficiency. They will have a deeper understanding for students of core STEM subjects to allow them to, as Dr. Leitner mentioned, continue with the 5e learning model as well as the 5c's for the profile of a Virginia graduate. We want to make sure that they are enhanced in their critical thinking, which is across the board and cross-curricular. Old Dominion will help us with their sustainability of our teacher development and programming, as well as developing more of

[02:02:14] a teacher-educator pipeline. As Dr. Simerski mentioned, we're already in progress with that, but we've also partnered and have identified for the last two years STEM leads in each building who receive professional development monthly, bimonthly, from 21st Ed. And we've really been able to build the instructional capacity of our teachers there. So all of these efforts will increase our research focus on areas, our teacher development and building capacity

[02:02:48] for our learning. We have three goals. These are the metrics, simple but complicated. By the end of each lab school year, 2025 through 2029, the STEM Academy Lab School student testing in grades three through five will successfully meet and exceed that word is not, those words are not in there, but that's the goal. The minimum pass score on all SOL assessments. By the end of each lab school year, the STEM Academy lab students will successfully

[02:03:27] meet and exceed the minimum pass score on all division generated, VDOE required, project-based learning, and other assessments. And 100% of our STEM Academy lab school teachers and staff will practice and implement professional development strategies in which they were trained and are embedded within the STEM Lab Academy School. We thank you for this opportunity

[02:03:57] and welcome any questions or comments. Any questions? Mr. Hanson? Thank you for this presentation, but also previously with the committee. And just thank you also just for vision of the elementary school approach here um could you maybe just help us a little bit too i know on the committee consideration there's a little bit of reference to booker t washington and just the physicality of how it's situated and how you're going to be drawing students from the

[02:04:31] community um into the school area but just a little bit maybe just even a thought process behind that and using that as the hub of the project one booker t washington thank you mr Hanson, Booker T. Washington is one of our smaller elementary schools, and it has about 340 students in a 500 to 600 capacity building. So location in the inner city of Suffolk, it is actually in what's considered the downtown area, has a lot of rich history in the school, but it has the

[02:05:08] building capacity to encompass the additional classes, the lab, and all of the extended learning there. So we really did want to focus in on where in some places we're over capacity, where we have the ability, the capability physically to do it. We're also, as far as bringing students there, we're going to be using the lottery system as designed in the lab school application process and required. And we'll be working with Old Dominion University to create

[02:05:40] that lottery system. And for the students there, they will be part of that building. And so they will receive the same transportation, food and nutrition services, all things that are related to our students. But the location is based on the size, the physical location and rich history of the building. Dr. Northern, what curriculum will you be using? Our plan is to use Project Lead the

[02:06:16] Way. They have an elementary curriculum. We currently utilize that at two of our high schools, which are nationally recognized, I might add. So we plan to adopt the elementary Project Lead the Way curriculum to utilize for this. And you have already, did I hear you say that that's already in place in some of the high schools? Yes, ma'am. Okay. And you have the money already to adopt project lead the way or do you have to get grant funding to do that we are i just i don't know if you want to speak to that but we already have that in place from our planning grant okay excellent just for my colleagues it's a very reputable stem um hands-on curriculum so kudos for choosing that

[02:06:52] one thank you miss ashton i just also want to applaud the effort um as a middle high school stem school leader thinking about what our kids if they receive this education at elementary school the amount of progress and excitement um and just retention and college-going readiness that we could start early is commendable i also want to echo project lead the way is an incredibly rich rigorous curriculum and the alignment for you know elementary 312 i think it's just going to

[02:07:28] accelerate the learning um and the last is just kudos to the partnerships like all i kept hearing with just partnerships, ODU, Suffolk, the school teachers. So I'm just incredibly excited to have the proposal, see it come into life. So thank you. Thank you. Ms. Holton. Yes, I thank you all for being here and presenting this exciting project. And I join my colleagues and especially, I'm especially excited, yes, to have an elementary school in our portfolio of lab schools. Very

[02:08:01] exciting. And I love that clearly embedded deeply in your plans as outlined in the metrics is the focus on the teacher development. This really is about improving student learning, but also improving how we train our teachers. So kudos on all of that. Help me understand. So this is a school within a school. Is that right? So you're going to be serving some but not all of the students at Booker T. Washington? And how's that going to work? We are going to allow the lab school lottery

[02:08:41] process to drive the enrollment. We do want the majority of our students to be from Booker T. Washington. That would be ideal. However, we're going to be true to the lottery process and create in partnership with old dominion that criteria that does recognize based on the criteria those students who need it there will be a wait list as is outlined in the lottery school application but the focus we would love to expand this after the lab school process to the entire school and

[02:09:18] make it a full school stem lab academy so starting as a school within a school building a separate school within the school building but your goals eventually would be to take the process to the whole school yes and i want to clarify i know for coding and as we do the data for the building those students will be booker t washington students however they will be identified in those six classes, K through 12, for our monitoring and assessing and ensuring not a

[02:09:53] tracking system, but making sure that what we are proposing and doing is impacting those students. Thank you. And I should know this because I know it's embedded in our lottery process, But do the current Booger T. Washington catchment area students get any, do the criteria allow them to go to the head of the line in the lottery or are they just in with anybody else?

[02:10:21] In the lottery process, we have to be fair and consistent. We do want to target our students, but we have to go with the exact process as outlined in the lab school. And so I do not want to say that those students would have any higher ranking or priority.

[02:10:43] However, we really do want our students at Booker T. Washington Elementary School to be at the forefront, but we will use the true lottery system. So they don't get any extra waiting in the lottery, but obviously they'll be aware of the school marketing, recruitment, et cetera.

[02:10:59] Yes, we will market strongly with our team at Booker T. Washington and within our community. And then you're starting in fall of 25, I assume? Yes, this will be once approved, if approved, it will be our planning year, 24-25, and we will launch fall 2025.

[02:11:19] And then the goal is to serve 120 students per year. Is that right? Yes, 20 per class per grade level, so 120 per year. And so that's K through 5, 20 per class per grade level.

[02:11:36] And then what is the total amount of money you're seeking from the lab school fund? I want to turn it over to Dr. Sanzo, who has the full budget. Sure, yeah, and I apologize, it's great to go a little bit on our end.

[02:11:54] If you don't mind repeating the question, I couldn't quite follow that from the last question. Just what is the total budget that you are requesting, hold on, what is the total budget you are requesting from the lab school fund, the total amount?

[02:12:16] Sure, I need to close up quickly. I'm giving one, I don't have that spreadsheet. Dr. Armstrong, are you prepared? Dr. Armstrong in the room is indicating he has the answer.

[02:12:29] Yeah, it's 4.5, so we're requesting, so $1.2 million for the startup, so the $200,000 planning grant and $1 million for the year zero, so then we'd be looking at $3.3 million for the 120 students coming in, so $1,400 year one, $1.5 essentially operating.

[02:13:03] I think those are the numbers. Dr. Arshon, is that what you're looking at? He's indicating agreement. Thank you. Yeah, yeah. It's comparable to the request that we had for Newport News with that particular one.

[02:13:19] The interesting element that you all know when we look at a K-5 school is not only are we looking at providing support to teachers and administrators, we also have the additional need to have instructional assistants that are coming into the classrooms but takes into account that budget that is um comparable to what we had already requested with the number of students actually i mean if you look at it for students actually less than what we were asking for the new previous one thank you dr siebert i know you had a question

[02:13:56] or comment uh comment and encouragement so thank you very much and uh appreciate what you're doing here as i reviewed your application listening presentation today you have something that's built in that i would encourage you to lift up and that is you're creating an end for accountability you concluded with the fact that you want to look at our traditional decades-old approaches of accountability and commonwealth we also began with the fact that because this is going to be a demonstration project-based phenomenon-based problem-solving approach that you're going to have the opportunity for the community at large not just parents to come in and see and that it's

[02:14:33] just important that we bear in mind that often we think of accountability in a traditional sense in terms of test scores and those are very important but there's very real very genuine accountability in project-based learning environments that everybody gets to come in and see and so uh i that's an n for accountability and as we in the commonwealth continue to think and work on accountability i would encourage you get all this built in i'd encourage you to lift it up thank you thank you dr northern yeah how much if i could add to that as you think about the accountability element we are looking at is dr grant and others have mentioned alignment across the three lab

[02:15:13] schools so the elementary level just like the middle school we're looking at this expositions and partying that i i'm going to take your language and add that to what we've already um you know included in the grant because i think you've raised a good point that we've talked about Dr. Northern, I'm just wondering how much time I know different project lead the way schools organize instruction differently. So how much exposure will the kids be getting to the curriculum?

[02:15:48] What does that look like time-wise a week? Certainly. So we have not purchased it quite yet, but we've met with the committee. And so I started with an alignment document. So while it is aligned, thank you so much. We had to ensure we're accounting for all SOLs.

[02:16:03] So they'll be engaged weekly within the curriculum. So we're purchasing all the curriculum that's solely aligned to the Virginia Standards of Learning. So weekly, they will be engaged in the Project Lead the Way. I don't have an hourly breakdown based on the purchase of what we get and all of the supplies and materials. However, what was accounted for is that every single week, it'll be a daily process.

[02:16:30] So they'll be engaged in it starting with Monday with the proposed design. So whatever the problem is, and then each day it's built out all the way through that Friday. It's almost like a week-long curriculum that they'll be exposed to it. Further questions, comments?

[02:16:48] Dr. Hanson, do you have anything? Okay. All right. Thank you so much. We appreciate it. Thank you. Let's see. As we move on, what I'd like to ask now is if there is a motion to waive first review and approve the ODU Suffolk Lab application for five years with annual reviews in accordance with the board's bylaws with total startup and operational lab school per pupil funding from the college partnership laboratory school fund to be determined no later than june

[02:17:26] 21st 2024. they said that okay guidelines and bylaws boards guidelines and bylaws okay any Any discussion?

[02:17:49] Yes, Mr. Rotherham. Further comments or discussion?

[02:18:17] Ms. Ashton, Dr. Northern, Mr. Hanson, Dr. Sievert, Ms. Holton. and share votes aye. Okay, thank you so much. Thank you. Okay, Dr. Armstrong. Madam President,

[02:18:57] board members, thank you. I'd like to now provide an overview of the George Mason University, Shenandoah Valley Rural Regional College Partnership Laboratory School for Data Science, Computing, and Applications, also known as DSCA. This is a partnership among George Mason University, Clark County, Fauquier County, Frederick County, Page County, Shenandoah County, Warren County, Winchester City Public Schools, as well as Laurel Ridge Community College and the Governor's School at Mountain Vista. I would like to again thank the members of the College Partnership Lab School Standing Committee for their time, feedback, and commitment to lab schools, and you will find

[02:19:33] cover sheets from the OAG and VDOE, as well as consistent budget templates in each application reflecting the additional efforts taken on behalf of the applications before you today for consideration, including this application from GMU DSCA. The GMU DSCA Lab School proposal was presented to the Lab School Standing Committee on May 6, 2024, and was unanimously approved on first and final review to be forwarded to the State Board for your consideration today.

[02:20:01] Joining me today from George Mason University are Dr. Padmadian Sashair, Professor and Director at George Mason University, and Dr. Deb Crawford, Supervisor of Mathematics and World Language Instructional Services Department at Frederick County Public Schools. Upon completion of my report, I will invite the GMU team to provide a brief overview of their proposal and to introduce any other special guests representing the GMU DSCA Lab School. The proposed lab school named GMU DSCA is an innovative interdisciplinary lab school model that focuses on providing career exposure to students to real-world problem solving in collaboration with higher education professionals, pre-service college students, and industry experts. Students will remain enrolled

[02:20:41] in their homeschool division while being provided with several options to engage in career exploration, including dual enrollment opportunities at Laurel Ridge Community College. The proposed lab school will serve 50 students from grade 10, 50 students from grade 11, and 50 students from grade 12 who will have the opportunity to engage in the SCA through five common experiences, which include interdisciplinary coursework, where students will be exposed to data-focused coursework from various aspects of STEM that applies to real-world workspaces, research and creative experience, in which students will work on specific student-led research projects where he or she will work closely with the faculty across the Commonwealth and other researchers, helping

[02:21:21] them to earn college credit through George Mason University, design thinking and entrepreneurship experience in which students will learn to solve problems using a human-centered approach to problem solving, and in the process, learn to build viable business models and a work-based learning experience component in which students will be mentored by big data business and industry partners through a choice of work-based experiences, as well as a global and multicultural experience in which data collaborations around global challenges are engaged where every student will interact with students from developing countries. All students will also have the opportunity to earn directed online data and analytics-focused certifications through companies

[02:22:00] such as Google and Amazon. Instruction will be administered in a hands-on environment and includes simultaneous teaching by Frederick County Public School staff, George Mason University faculty and pre-service students, Laurel Ridge faculty, and industry experts. The lab school would address multiple desirable outcomes for students, future teachers, and regional schools with hiring needs, including students engaging in high-level application of real-world problem solving while working closely with a team of experts from both industry and the higher ed partners. GMU's BSCA will provide a space for innovation where educators, industry, and pre-service professionals can explore best practices in teaching and learning, and students

[02:22:38] will engage in coursework aligned with personal interests and regional market demand industry and business, streamlining the pipeline for high-wage, high-skill, high-demand employment in the region served by the lab school. The VDOE received the DSCA application on April 2, 2024, and in the intervening months, we have collaborated with the applicant as follows. VDOE staff provided technical assistance to the DSCA team in preparation for standing committee review. Prior to their application submission, technical assistance calls were held between VDOE and DSCA team going back as far as August 17, 2023, shortly after I came on board, November 2, 2023, November 14, 2023,

[02:23:17] and March 18, 2023. Following submission of their application, additional technical assistance was provided in the first round of counseling from this subject matter expert review was provided to the applicants during the week of April 16. A revised application was submitted to the VDOE on April 18, and an additional application responsive to OAG and VDOE review was provided to the VDOE on April 25th for completion of the report that went to the College Partnership Lab School Standing Committee. Follow-up feedback was provided to the applicants and the team presented to the Standing Committee on May 6th, responded to their questions, and was approved by the Standing Committee on May 6th unanimously to proceed to today's special meeting of the Board of Education

[02:23:57] for consideration for approval. The GMU-DSCA Lab School application was posted for public comment following the May 6th standing committee meeting on May 6th and remained available until May 20th, and we did not receive any comment. On behalf of the VDOE, I submit that the George Mason University DSCA application is compliant and ready to implement, and I recommend to the board that the lab school be approved and subsequently begin the contract approval process. Again, here to share more about the GMU DSCA lab school, I am joined by Dr. Sashair, as well as Dr. Crawford, who will provide additional information and introduce any other guests who are here on behalf

[02:24:31] of DSCA. Dr. Crawford and Zeshair. Morning. Thank you for inviting us here. My name is Padu Zeshair and I'm a professor and director at George Mason University and once again we are very honored to be here in front of distinguished members of the Board of Education and here I'm

[02:25:07] with my collaborator Dr. Deb Crawford who's a supervisor of mathematics and world languages of Frederick County Public Schools, and also an instructor at George Mason University. Together we have led a Shenandoah Valley Regional Computer Science partnership with 11 partners in the valley, with GMU serves as the lead higher ed, and Frederick serves as the lead school division, and we both also have served over the past five years as the Virginia High School Data Science, higher ed, and K-12 leads respectively on the standards writing

[02:25:41] implementations so rural and rural fringe divisions such as those in shenandova valley are often overlooked uh part of the complex american education system even though about 10 million

students one in five nationally attend a rural or rural fringe school so targeted facilities and investments such as the videoe college partnership lab school opportunities could pay dividends in improving rural education so once again thank you for inviting us uh to this to present uh today the shenandova valley rural regional uh college partnership laboratory school that will

[02:26:20] focus on dsca which is data science computing and applications this lab school will be led by gmu as a and it's a collaborative project with seven uh partner divisions and one governor school in in the Shenandoah Valley with Frederick County as the lead Kate Wall partner.

[02:26:37] GME will also leverage the expertise of faculty from the George Mason University College of Science, the Complete Center, which is a joint center between College of Science and College of Education and Human Development, the Smithsonian Mason Conservation School of Conservation at the Front Royal Campus for George Mason and Laurel Ridge Community College to engage students and teachers in this GMUD SEA lab school.

[02:26:59] Thank you so much for the opportunity to bring a data science and Computing Applications Lab School to students in the Valley. The lab school will employ a hub and spoke model that will be led by GMU, working with VDOE and Frederick County Public Schools as the hub where the physical lab school will be located, along with the lab school opportunities provided through GMU, SMSC, and Laurel Ridge Community College for students, teachers, and parents from all over the valley from partnering divisions serving as the spokes. Our mission is to

[02:27:35] create innovative pathways designed to help students and teachers from rural and rural fringe school divisions add data science and computing applications expertise to their skill set and make informed data-driven decisions that contribute meaningfully to their communities. Our vision is to create a data literate next generation workforce empowered with the skills knowledge and mindset necessary to thrive and solve problems in a data-driven world to bring about positive

[02:28:09] change think about it in the past two years over 90 of the data that has ever existed was created and most of it is unstructured data requiring new methods for analysis and use for decision making Excellent. So the lab school will be hosted physically at Double J Harvard Center in Frederick County Public Schools, and it'll serve students each year as a full lab school. Lab school students

[02:28:45] and the partner division faculty teachers will have the opportunity to take dual enrollment coursework and professional learning opportunities through the regional higher education partners, including George Mason University, Smithsonian Conservation Center, Laurel Ridge Community college the proposed lab school will serve 50 students from uh each of the grades 10 11 and 12 who will have the opportunity to engage in data science computing and applications through five common experiences so these experiences are actually uh foundational competencies identified by the national academy of engineering grand challenge scholars program these include uh

[02:29:24] an interdisciplinary coursework experience a research and creative experience design thinking and entrepreneurship experience, work-based learning experience, and global and multicultural experience. And also, with Governor Youngkin's executive order on AI issued in January 2024, the need to prepare the next generation workforce on competencies for data in the age of AI becomes even more important. Which is the second data science course they'll take in the lab school.

[02:29:51] This day in the life slide that you see up there shows sample schedules for 11th and 12th graders from the industry-aligned data hub, which serves as the center of the lab school. Note that travel to and from the lab school from the base schools allows opportunity for lab school students to engage in after-school clubs and sports.

[02:30:12] The schedule shows how two elective periods per day allow for customizable workplace learning options, graduation requirements both for advanced studies, diploma, associate's degree, and university research for every student in the lab school with mentorship from university faculty with data science computer science and scientists such as those from the smithsonian mason school of conservation which is right in our backyard in front royal and the smithsonian is hugely excited about working with us this is something they've long wanted to do in the valley students in the

[02:30:50] lab school will also have the opportunity to engage throughout the curriculum in a no code low code and high code environment in terms of their experiences the primary difference between those lies in the level of coding expertise required the degree of customization and control offered and the target audience pbl and the data cycle underlie all of the curriculum that we are using and integrated into our instructional model as well thank you and you can see faculty

[02:31:29] from ODU faculty from Shenandoah University teachers from different school divisions so so the lab school is a data-focused school offering an emphasis on advancing data science computing and applications through innovative research and education through this collaborative partnership with multiple universities as well the proposed lab school will help build on successful best practices you know these regional partners have already established through three uh virginia department education advancing computer science education grants that helped to

[02:32:04] enhance the regional partnership and we were also able to develop a micro pilot lapse component in these grants within this project which allowed high school students over the last three years to engage in capstone projects earning college credits in the areas of data science computing and applications and through this we are able to engage faculty from multiple universities across the commonwealth as well as teachers from all the partnering divisions.

[02:32:31] So this framework that you see up on the slide is not hypothetical. We have been engaged in student research projects for three years in the valley and we have a showcase tomorrow night at Shenandoah University for our students and you can see that model has the student at the center and their high school teachers along the bottom although we have more of them and then their university mentors are at the top all supporting their student the students on their student-led research projects based on the expertise required for the project data and computing provides a significant

[02:33:05] competitive advantage in today's society with a need to expand the talent pool to address workforce needs we need to create new programs designed to help our students add dsca expertise to their skill set also the recent national council of teachers and mathematics released a position statement. You may have seen it on data science and the value of that high school course in their curriculum as a credit for graduation. In the data science lab school, students will be getting not one but two data science courses, second at the college level, two computer science courses at the

[02:33:41] college level, and two advanced science courses involving data applications. Again, one of them at the college level and working with those scientists at the Smithsonian. So very cool opportunities for students that are not available at the base school. All students will also have the opportunity to earn

directed online data analytics focused certifications through companies such as Google and Amazon. Students in the lab school also have the opportunity to earn an associate's degree from Laurel Ridge Community College. We have the pathways set up. College credit from George Mason University, as I mentioned before, and industry certifications from partnering

[02:34:18] organizations that you see up here and i was fortunate to be one of the authors on the from representing virginia on the nctm statement on high school data science and to sustain uh the next slide please so uh the total budget we asked was for 6.25 million and to sustain uh the proposed college partnership school model various resources will be essential including financial resources political capital as well as staff talent and interest so it's very important for long-term viability and as a former program director at the nsf i know how important it is to raise funds so

[02:34:55] and so it's not just the initial implementation but also ongoing operations maintenance and growth so starting year two of the lab school we plan to bring our experience to apply to secure grants from the nsf nih videoe nasa as well as foundation funding we actually met with gates on saturday and simon's monday yesterday so simon's so there are foundations that are also looking at data science as something that they can actually contribute to we'll also be actively pursuing partnerships and funding from non-profit foundations such as discover e that

[02:35:29] runs the future city competitions battle foundation broadcom foundation both of which are funded gmu for several of their stem programs as well as for-profit organizations such as including local businesses and industries we'll also try to pursue congressional earmarks for our lab school each year which are opened up as community projects in the house and congressionally directed spending in the senate i am the only hired member that serves on a stem advisory board as well as the workforce development board so it gives us the opportunity to learn about these

[02:36:08] congressional funding that is available each year and we'll also follow up with the governor schools that we both have been involved with there's a Prince William governor school model that George Mason has been working with and understanding best practices from that that we can use for our lab schools and so we've talked a lot about the opportunities for our students but what about our teachers we are excited that the data science and computing applications lab school will also serve as a teacher hub in the commonwealth for data science and computing so how will we do this

[02:36:44] one of the novel innovations in our proposed lab school will be the development of digital badges and micro credentials in data science and computing for teachers and leadership that will be developed in conjunction with gmu and laurel ridge community college these badges and micro credentials will serve as tangible evidence of skills and knowledge that teachers are acquiring also the targeted approach can be more efficient and cost-effective most importantly these can be accessible customizable and stackable both of us dr seshire and i have experience working with

[02:37:18] vascd on creating micro credentials for data science and the vascd the virginia association for supervision and curriculum development we developed micro credentials there around various community engagement aspects of the high school data science curriculum. The roadmap that you see in the top right shows teachers as designers and learners within the lab school environment.

[02:37:43] The lab school also seeks to serve as that go-to data science and computing hub that I mentioned for teacher professional learning across the Commonwealth. We have a data science institute. 115 teachers have already signed up slated for June 25th and 26th at George Mason University.

[02:38:00] These are teachers in the Commonwealth who want to learn how to teach data science since we have new standards that were just approved two years ago. So we are already serving as this hub and really would like to scale that up. Teachers in the lab school, but also from the partnering divisions, may participate in these opportunities to upskill data science and computing skills based on a pre-assessment.

[02:38:25] Attendance at research courses with their students, we're already doing that. connecting with university experts in their fields through student research projects and business and industry partners, Google data analytics certificate support, classroom visits, monthly cohort support groups, and lesson study around topics of interest.

[02:38:43] Let me give you just a quick example. The 115 teachers coming to the Institute in June, they're going to get a QuantHub personalized dashboard login so they can do a pre-assessment of where their skills are, and then they can work on those modules that they would like to work on.

[02:39:00] That's an example of how customizable this is for teachers to do. Yeah. And and, you know, through this partnership, we are very confident that the JMU College Partnership Lab School will help facilitate pathways for data science fostered through not just high school, but various stages of education, you know, through community college, through undergraduate studies and all the way to.

[02:39:24] So we are already seeing these kids going to take bachelors in data science or minors, uh you know so uh with other majors that they would like to see and so we have a lot of activities uh like even for parents uh to be getting to be involved we call it super data saturdays which will be at the smithsonian mason school of conservation and extending this experience through activities uh later in college as well so and once again we want to close with the importance of uh you know engaging rural and rural french divisions in in in data in the age of ai so i think it's very important that we do it early and i think this is how we can impact the next generation

[02:40:04] thank you something very important thank you very much thank you does the board have any questions mr hansen just again thank you and um i just uh i'm just thrilled with the proposal just obviously we talked about this in the lab committee meeting but uh both the geographic uh needs and uh work of george mason and the collaboration with the school system i just um and for that region it's just critically important so thank you um i also just when you're talking about the teachers

[02:40:42] i think in my two years on the state board um in our your presentation when you had the three students and the three teachers with them it was probably the most profound experience that i've had to see the um each of those three students were just so incredibly impressive and but also the uh the love the support the professional um partnership with their faculty was just uh so clear and it just was uh to me just a could not sometimes i'm a visual learner but it was just

[02:41:20] such um so clear of what can be achieved through this project so i just i wish the entire board could have witnessed those three students and the three faculty because it really was both and i think to your point of the you know teachers uh training the teachers and um you know that is so uh critical here but i just wanted to thank you for uh the thoughtfulness and the collaboration that you've put together here just looking forward to really helping the region in a very measurable thank you very much for that comment and uh just with uh we were surprised one of the

[02:42:00] students declared her major should be data science and she was just entering and that is the day we found out also and she's going to be presenting tomorrow so thank you again our students stole the

show as they should and we wish they were here today we had about eight other people yes further comments or questions dr northern so i'm sure you're probably aware of the debate within the mathematics field relative to calculus versus data science. You know, folks, there are many folks that argue that taking, you know, algebra to pre-calculus and calculus is a better route for

[02:42:37] STEM, for kids who are interested in STEM than data science. And then other people say, no, you shouldn't be one or the other, you know, these should be more encompassing. So I guess, where do you stand on that sort of, you know, for certain kids, it's better to take calculus and data science, or how do you guys make sense of that debate in the mathematics field yeah i'll speak from the higher so yesterday's meeting in new york by simons foundation simons is the billionaire mathematician he organized about 25 organizations to come together i am the chair of the u.s national academy's commission for mathematics instruction this is exactly what was brought up and i was

[02:43:14] i was able to present the case for data science and uh you're right exactly so you know but uh are we trying to so the discussion was are we trying to get all kids to do calculus after algebra 2 or pre-calculus or are we trying to find kids with or are we trying to build competencies which will help a kid that is going to do restaurant and hotel management that can predict which coffee sells the most at 2 pm on a Wednesday Starbucks will hire them for 100k so it's about you know predicting which one so that that aspect of so one of the things that we have

[02:43:48] actually developed in this is a cycle of acquisition and collection of data uh interpretation of processing and and cleaning of data and uh interpretation and visualization of data modeling and analysis of data and uh lastly prediction and validation of data each of those steps has mathematics in it so people are now starting to understand the importance of data science as not being mathematics but it's mathematical foundations as not being statistics it's statistical reasoning as not being computer science it's computational thinking so if i think

[02:44:25] of a triple circle data science sits at the hub addressing health or some domain and so now you're so it is only the mathematics community but it's right now they're starting to discuss the more awareness they are getting it it's becoming better so for example my presentation yes it was helpful to many of them to get clarified otherwise you know so there are lots of messages and uh there's also national organizations called data science for everyone who was who were there so you are also trying to uh so we are uh trying to promote this message and so one very important

[02:44:59] point your uh comment is virginia has been the first state to approve uh uh the board of education approved uh standards of learning no other state has done it and we probably have heard lots of things going on in california the reason it worked is because it was it was built you know bottom up with higher ed capable supervisors vdoe everybody coming together and actually building the standards of learning and then going up so we are at that point in the mathematics community to see like all these kids that are coming in they need to get ready so you know so one part of my job is

[02:45:36] to also help align all the institutions to help them understand given this like 10 different data data science classes inside a university so I don't think so because they all will be expected

[02:46:13] take a math placement exam and so you know their the ability to be able to place into some class will be you know each university has their own standards so by doing data science though they will have an opportunity that normal students may not have is to the ability to get a major double major a minor and they may not go into a major in mathematics or engineering they could get a minor and so I'm

I'm already started like the girl that surprised us at the standing team meeting that she's going to do data science major and a computer science minor so i think these are the types of things that we're going to be seeing a lot actually so great question

[02:46:52] and i hope you don't mind if i add a little bit from the student standpoint at the lab school so students have different pathways in the lab school that they can choose based on their interests and their career goals and we have students who will come in and work on a data analytics certificate, and they can go straight to work, a really, really good job, you know, 60,000, right?

[02:47:17] Simultaneously, they can work and then go to college, and the work will pay for them to go to college, or they're going straight to work. It'll give them that workforce skill set. And as you know, data science and computing underlie AI, and whoever can figure out AI wins. So this is needed by our students. Then we have students who have taken calculus, and they're earning an associate's degree at Laurel Ridge Community College. They've already completed two years of college by the time they leave the lab school. They're doing university research. So they're

[02:47:51] publishing. So it's not one or the other. Then we have another group that's doing internships after typically a data analytics certificate or credential. And so they're going, they may go straight to college, but have that internship piece where possibly some of that college could be funded. And we are looking locally at businesses who are interested in doing that.

[02:48:13] So it's not just one or the other. The Ivies have backed off on calculus. And it's kind of the idea that the STEM workforce needs a more diverse view of what mathematics is. And really, data science is really at the heart of mathematics, statistics, computer science, and the domain of any data set and so that's it's it's really not just one answer your question is great though great

[02:48:46] question right actually yeah and right so the that calculus may not be the entire calculus that that kid needs to take but there are certain parts like you know what is the best is a very important concept here in the data cycle so what's the best car you know what's the cheapest i should pay all that takes uh elements from calculus so you know so that's how you know some of them are

[02:49:21] getting convinced to understand what the content is and i also want to say because i love calculus too um i want to say it's not a matter of data science or calculus it's a matter of when when would you take calculus and when you know you see what i'm saying data scientists who who have that college degree in data science they have taken calculus several calculus courses so kids can get really excited and into data science which brings them into and keeping them in mathematics and on that stem track of the university which if you've looked at the numbers particularly females is not

[02:49:59] great and so if you if you think this is a kind of a pathway they'll be fired up to take calculus in a couple of years. Or not. Any further questions? Ms. Holton? Just welcome, Dr. Sechere, Dr. Crawford. Great to hear your presentation. Just so you know, I will recuse myself when we come to the vote on this matter because of my affiliation with George Mason University.

[02:50:25] Thank you. Other questions? Okay. Thank you all. So that being said, is there a motion to waive first review and approve the GMU Frederick Lab School application for five years with annual reviews in accordance with the board's guidelines and bylaws with total startup and operational lab school per pupil funding from the college partnership lab school fund to be determined

[02:50:52] no later than June 21st, 2024. Any discussion? Ms. Ashton, Dr. Northern, Mr. Hanson, Mr. Rotherham, Dr. Siebert, and chair votes aye. Oh, I'm sorry, Ms. Holton, I know you have to recuse

[02:51:15] to make that official. Thank you. All right. Thank you so much. Dr. Armstrong, do you have anything in addition to your written report? Thank you. With nothing else on our special call meeting agenda, I will entertain a motion to adjourn. All right. All adjourned. Thank you.



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